

Assignment-2 (NLP)

Name: Bhavaneesh.A

Batch: 013

Roll Number: 2403A52350

```
import nltk
from collections import Counter
import spacy
```

```
# Downloading NLTK neccessary things
```

```
nltk.download('punkt')
nltk.download('averaged_perceptron_tagger')
```

```
... [nltk_data] Downloading package punkt to /root/nltk_data...
[nltk_data] Package punkt is already up-to-date!
[nltk_data] Downloading package averaged_perceptron_tagger to
[nltk_data] /root/nltk_data...
[nltk_data] Package averaged_perceptron_tagger is already up-to-
[nltk_data] date!
True
```

```
# Load Spacy Model
```

```
nlp = spacy.load('en_core_web_sm')
medical_text = """
The patient reported a sudden headache and nausea.
Blood pressure was elevated, and heart rate was irregular.
The doctor prescribed aspirin and advised rest.
Follow-up tests confirmed improved condition after treatment.
"""
```

```
# NLTK Tokenisation
```

```
sentence_nltk = nltk.sent_tokenize(medical_text)
print(sentences_nltk)
```

```
word_nltk = nltk.word_tokenize(medical_text)
print(words_nltk)
```

```
['\n\nThe patient presented with acute chest pain and shortness of breath.', 'Electrocardiogram indicated signs of myocardial infarction.', 'Immediate administration of aspirin and nitroglycerin was advised. Blood pressure was elevated, and heart rate was irregular. The doctor prescribed aspirin and advised rest. Follow-up tests confirmed improved condition after treatment.\n']
['The', 'patient', 'presented', 'with', 'acute', 'chest', 'pain', 'and', 'shortness', 'of', 'breath', '.', 'Electrocardiogram', 'indicated', 'signs', 'of', 'myocardial', 'infarction', '.', 'Immediate', 'administration', 'of', 'aspirin', 'and', 'nitroglycerin', 'was', 'advised', '.', 'Blood', 'pressure', 'was', 'elevated', 'and', 'heart', 'rate', 'was', 'irregular', '.', 'The', 'doctor', 'prescribed', 'aspirin', 'and', 'advised', 'rest', '.', 'Follow-up', 'tests', 'confirmed', 'improved', 'condition', 'after', 'treatment', '.\n']
```

```
# Spacy Tokenisation
```

```
doc = nlp(medical_text)
sentences_spacy = list(doc.sents)
words_spacy = [token.text for token in doc]
```

```
print(sentences_spacy)
print(words_spacy)
```

```
[
The patient reported a sudden headache and nausea.
, Blood pressure was elevated, and heart rate was irregular.
, The doctor prescribed aspirin and advised rest.
, Follow-up tests confirmed improved condition after treatment.
]
['\n', 'The', 'patient', 'reported', 'a', 'sudden', 'headache', 'and', 'nausea', '.', '\n', 'Blood', 'pressure', 'was', 'elevated', 'and', 'heart', 'rate', 'was', 'irregular', '.', 'The', 'doctor', 'prescribed', 'aspirin', 'and', 'advised', 'rest', '.', 'Follow-up', 'tests', 'confirmed', 'improved', 'condition', 'after', 'treatment', '.\n']
```

```
# NLTK POS tagging
```

```
pos_nltk = nltk.pos_tag(words_nltk)
print(pos_nltk)
```

```
[('The', 'DT'), ('patient', 'NN'), ('presented', 'VBN'), ('with', 'IN'), ('acute', 'JJ'), ('chest', 'NN'), ('pain', 'NN'), ('and', 'CC'), ('shortness', 'NN'), ('of', 'IN'), ('breath', 'NN'), ('.', '.'), ('Electrocardiogram', 'NN'), ('indicated', 'VBD'), ('signs', 'NNS'), ('of', 'IN'), ('myocardial', 'NN'), ('infarction', 'NN'), ('.', '.'), ('Immediate', 'JJ'), ('administration', 'NN'), ('of', 'IN'), ('aspirin', 'NN'), ('and', 'CC'), ('nitroglycerin', 'NN'), ('was', 'VBD'), ('advised', 'VBN'), ('.', '.'), ('Blood', 'NN'), ('pressure', 'NN'), ('was', 'VBD'), ('elevated', 'VBN'), ('and', 'CC'), ('heart', 'NN'), ('rate', 'NN'), ('was', 'VBD'), ('irregular', 'VBN'), ('.', '.'), ('The', 'DT'), ('doctor', 'NN'), ('prescribed', 'VBD'), ('aspirin', 'NN'), ('and', 'CC'), ('advised', 'VBN'), ('rest', 'NN'), ('.', '.'), ('Follow-up', 'NN'), ('tests', 'NNS'), ('confirmed', 'VBD'), ('improved', 'JJ'), ('condition', 'NN'), ('after', 'IN'), ('treatment', 'NN'), ('.', '.')]
```

```
# Spacy POS tagging
```

```
pos_spacy = [(token.text, token.pos_, token.tag_) for token in doc]
print(pos_spacy)
```

```
[('\n', 'SPACE', '_SP'), ('The', 'DET', 'DT'), ('patient', 'NOUN', 'NN'), ('reported', 'VERB', 'VBD'), ('a', 'DET', 'DT'), ('sudden', 'ADJ', 'JJ'), ('headache', 'NOUN', 'NN'), ('and', 'CC', 'CC'), ('nausea', 'NOUN', 'NN'), ('.', 'PUNCT', '.'), ('\n', 'SPACE', '_SP'), ('Blood', 'NOUN', 'NN'), ('pressure', 'NOUN', 'NN'), ('was', 'AUX', 'VBD'), ('elevated', 'ADJ', 'JJ'), ('and', 'CC', 'CC'), ('heart', 'NOUN', 'NN'), ('rate', 'NOUN', 'NN'), ('was', 'AUX', 'VBD'), ('irregular', 'ADJ', 'JJ'), ('.', 'PUNCT', '.'), ('The', 'DET', 'DT'), ('doctor', 'NOUN', 'NN'), ('prescribed', 'VERB', 'VBD'), ('aspirin', 'NOUN', 'NN'), ('and', 'CC', 'CC'), ('advised', 'VERB', 'VBN'), ('rest', 'NOUN', 'NN'), ('.', 'PUNCT', '.'), ('Follow-up', 'NOUN', 'NN'), ('tests', 'NOUN', 'NN'), ('confirmed', 'VERB', 'VBD'), ('improved', 'ADJ', 'JJ'), ('condition', 'NOUN', 'NN'), ('after', 'IN', 'IN'), ('treatment', 'NOUN', 'NN'), ('.', 'PUNCT', '.')]
```

```
# From NLTK
```

```
nltk_nouns = [w for w, t in pos_nltk if t.startswith("NN")]
nltk_verbs = [w for w, t in pos_nltk if t.startswith("VB")]
```

```
# From spaCy
```

```
spacy_nouns = [t.text for t in doc if t.pos_ == "NOUN"]
spacy_verbs = [t.text for t in doc if t.pos_ == "VERB"]
```

Toggle Gemini

```
print("\nNLTK Nouns:", nltk_nouns)
print("spaCy Nouns:", spacy_nouns)
```

```
print("\nNLTK Verbs:", nltk_verbs)
print("spaCy Verbs:", spacy_verbs)
```

```
NLTK Nouns: ['patient', 'chest', 'pain', 'shortness', 'breath', 'Electrocardiogram', 'signs', 'infarction', 'Immediate', 'administration', 'aspirin', 'nitroglycerin', 'ect
spaCy Nouns: ['patient', 'headache', 'nausea', 'Blood', 'pressure', 'heart', 'rate', 'doctor', 'aspirin', 'rest', 'tests', 'condition', 'treatment']
```

```
NLTK Verbs: ['presented', 'indicated', 'was', 'performed', 'showed']
spaCy Verbs: ['reported', 'elevated', 'prescribed', 'advised', 'Follow', 'confirmed']
```